

Rigid Analytic Geometry

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1 Introduction - Spectral Spaces and Spaces of Valuation Spectra

1.1 Introduction

Definition 1.1. Let $\Omega \subseteq \mathbb{C}$ be a domain. We define a sheaf of holomorphic functions as,

$$\Omega \supseteq V \mapsto \mathcal{O}(V) = \{f : V \rightarrow \mathbb{C} \mid f \text{ is holomorphic}\}.$$

One can equip $\mathcal{O}(V)$ this with the semi-norm such that,

$$\|f\|_{V,A} = \max\{|f(z)| \mid z \in A\},$$

where $A \subseteq \Omega$ is a closed and bounded in $\mathbb{C}$, namely a compact set A . With respect to this semi-norm, we define basis elements of the topology in the following way.

Proposition 1.1. The semi-norm induces the topology on $\mathcal{O}(V)$ whose basis elements of the neighbourhood of 0 are of the following form. For $f \in \mathcal{O}(V)$ and $\varepsilon > 0$,

$$\{f \in \mathcal{O}(V) \mid \|f\|_{V,A} < \varepsilon\}.$$

Definition 1.2. A ring is said to be Bézout if every finitely generated ideal is principal.

Remark 1.1. By Weierstrass product theorem, $\mathcal{O}(V)$ is not Noetherian but it is Bézout.

Proposition 1.2. The stalks of $\mathcal{O}(V)$ which is a ring of germs of holomorphic functions are Noetherian. This holds for holomorphic functions in several complex variables as well.

The goal of the course is to generalize these results to p -adic cases.

We will allow ground field K with an absolute value $|\cdot| : K \rightarrow [0, \infty)$ satisfying,

$$|0| = 0, |1| = 1, |ab| = |a||b|, |a + b| \leq \max\{|a|, |b|\}.$$

Definition 1.3. Let K be a field which is complete with respect to the map $|\cdot|_K : K \rightarrow [0, \infty)$ is an absolute value of K if we have the following conditions.

- i). $|1|_K = 1$,
- ii). $|0|_K = 0$,
- iii). $\forall a, b \in K, |ab|_K = |a|_K |b|_K$.

Furthermore, it is called ultra-metric if we have,

$$\forall a, b, |a + b|_K \leq \max\{|a|_K, |b|_K\}.$$

Proposition 1.3. Suppose we have an ultra-metric $|\cdot|_K$ on a field K . If K is complete with respect to the topology induced by the metric $d(x, y) = |x - y|_K$, then for a finite extension L/K , there is an extension $|\cdot|_L : L \rightarrow [0, \infty)$ of $|\cdot|_K$ such that L is complete with respect to the canonical metric obtained from $|\cdot|_L$.

Proposition 1.4. Let K be a field with an ultrametric $|\cdot|_K$. The completion $\widehat{K}$ of the algebraic closure $\overline{K}$ of K with respect to $|\cdot|_K$ is again algebraically closed.

Example 1.1. In Proposition 1.4, we can take for example $\widehat{K}$ to be $\mathbb{C}_p$ where $K = \mathbb{Q}_p$.

We will now give the definition of our basic objects.

Definition 1.4. Let $(K, |\cdot|_K)$ be a field which is complete with respect to the absolute value $|\cdot|_K$. We define the Tate algebra over K as

$$\mathbb{T}_m(K) := \left\{ f = \sum_{\alpha \in \mathbb{N}^m} f_\alpha X^\alpha \mid \begin{array}{l} \text{for all } \varepsilon > 0, \text{ there are only} \\ \text{finitely many } \alpha \text{ such that } |f_\alpha| > \varepsilon \end{array} \right\}$$

We define,

$$\|f\|_{\mathbb{T}_m} := \max\{|f_\alpha|_K \mid \alpha \in \mathbb{N}^m\}.$$

Proposition 1.5. Suppose K is algebraically closed in Definition 1.4. Then we have,

$$\|f\|_{\mathbb{T}_m} = \max\{|f(z)| \mid z = (z_i)_{i=1, \dots, m} \in K^m, |z| \leq 1\},$$

and such maximal is actually attained.

Remark 1.2. In p -adic settings, K^m is totally disconnected so assertion like the "identity theorem" for holomorphic functions becomes problematic.

In order to resolve the above obstacle Grothendieck introduced Étale topology.

1.2 Main Theorems of the Course

Definition 1.5. An element a is power-bounded with respect to a norm $\|\cdot\|$ if there is $M > 0$ such that for all $n \in \mathbb{N}$, $\|a\|^n \leq M$.

Definition 1.6. Let A be a topological ring. An element $a \in A$ is topologically nilpotent if

$$\lim_{n \rightarrow \infty} a^n = 0.$$

Definition 1.7. We define the following subset of a Tate algebra $\mathbb{T}_m$.

$$\begin{aligned} \mathbb{T}_m^\circ &:= \{f \in \mathbb{T}_m \mid f \text{ is power bounded}\}, \\ \mathbb{T}_m^{\circ\circ} &:= \{f \in \mathbb{T}_m \mid f \text{ is topologically nilpotent } \lim_{k \rightarrow \infty} f^k = 0\}. \end{aligned}$$

Proposition 1.6. $\|\cdot\|_{\mathbb{T}_m}$ is multiplicative. (i.e. $\|fg\| = \|f\|\|g\|$ and $\|1\| = 1$). In particular,

$$\mathbb{T}_m^\circ = \{f \in \mathbb{T}_m(K) \mid \|f\|_{\mathbb{T}_m} \leq 1\}.$$

and

$$\mathbb{T}_m^{\circ\circ} = \{f \in \mathbb{T}_m(K) \mid \|f\|_{\mathbb{T}_m} < 1\}.$$

From the definition of ultrametric, it is clear that $\mathbb{T}_m^\circ$ is a subring and $\mathbb{T}_m^{\circ\circ}$ is an ideal.

Proposition 1.7. $\mathbb{T}_m(K)$ is Noetherian and all ideals are closed in it. If $\mathfrak{m}$ is a maximal ideal, its residue field $k(\mathfrak{m}) = \mathbb{T}_m/\mathfrak{m}$ is a finite field extension of K (thus $|\cdot|_K$ extends uniquely to $k(\mathfrak{m})$).

Proposition 1.8. There is a topologically nilpotent unit, thus there are no proper open ideals.

Proposition 1.9. $\mathbb{T}_m(K)$ is regular of Krull dimension m .

Definition 1.8. A ring A is Jacobson if every prime ideal is an intersection of some maximal ideals.

Proposition 1.10. $\mathbb{T}_m(K)$ is a Jacobson ring. (The closed points of $\text{Spec } \mathbb{T}_m$ are dense in every closed subsets)

Notation 1.1. For a ring A , we denote

$$\text{Sp } A := \{\mathfrak{m} \in \text{Spec } A \mid \mathfrak{m} \text{ is a maximal ideal}\}.$$

Notation 1.2. Let $f \in \mathbb{T}_m(K)$ and $x \in \text{Sp } \mathbb{T}_m(K)$. We denote,

$$f(x) := f \bmod x \in k(x).$$

Proposition 1.11.

$$\|f\|_{\mathbb{T}_m} = \max\{|f(x)| \mid x \in \text{Sp } \mathbb{T}_m\}.$$

We define a basis element of the topology in $\mathbb{T}_m(K)$ as follows.

Definition 1.9 (Rational Open Sets). *Suppose $\mathbb{T}_m(K) = \langle f_0, \dots, f_m \rangle_{\mathbb{T}_m}$. The rational open set associated to the generators $\langle f_0, \dots, f_m \rangle_{\mathbb{T}_m}$ is*

$$\mathcal{R}_{\text{Sp } \mathbb{T}_m}(f_0|f_1, \dots, f_m) = \{x \in \text{Sp } \mathbb{T}_m \mid |f_0(x)| \geq |f_i(x)|, 1 \leq i \leq m\}.$$

Remark 1.3. *The Grothendieck topology enforces the quasi-compactness of these subsets.*

Additional points were introduced by van der Put in the late 1970s using a filter machinery equivalent to topos points. He also ? equivalence with ? valuation ? ? not

Berkovich in the 1980s constructed a multiplicative seminorm $\mathbb{T}_m \rightarrow [0, \infty)$ which are similar to $f \mapsto |f(x)|$ for $x \in \mathbb{T}_m$ or $f \mapsto \|f|_{\mathbb{T}_m}\|$. One then gets a compact Hausdorff space.

Huber constructed more general valuations and one gets a spectral space with a sheaf category equivalent to sheaves on $\text{Sp } \mathbb{T}_m$ (unlike Berkovich).

In the Berkovich machinery, one also has a point corresponding to

$$\|f\|_r = \max\{r^{|\alpha|} |f_\alpha|_K \mid \alpha \in \mathbb{N}^m\},$$

where $|\alpha| = \sum \alpha_i$ and $0 \leq r \leq 1$.

Remark 1.4.

$$\|f\|_1 = \|f|_{\mathbb{T}_m}\|, \|f\|_0 = 0.$$

Definition 1.10. *Let $0 \leq r \leq 1$. We define, for $f \in \mathbb{T}_m(K) \setminus \{0\}$,*

$$N_r(f) := (\|f\|_r, n)$$

where $n = \max\{i \in \mathbb{N} \mid \|f\|_r = r^i |f_i|_K\}$.

Example 1.2. *Consider the case $\mathbb{T}_1(K)$ and $f \in \mathbb{T}_1(K) \setminus \{0\}$ where*

$$f = \sum_{n=0}^{\infty} f_n T^n.$$

Then N_r defines a valuation, $\mathbb{T}_1 \rightarrow \Gamma \cup \{0\}$ where $\Gamma = (0, \infty)_{\mathbb{R}} \times \mathbb{Z}$ ordered lexicographically with the first factor being the most significant and $0 < \gamma$ for all $\gamma \in \Gamma$.

One has

$$N(a, b) = N(a)N(b), N(1) = 1, N(0) = 0, N(a + b) \leq \max\{N(a), N(b)\}. \quad (1.1)$$

When $0 < r < 1$ and $f \in \mathbb{T}_1 \setminus \{0\}$, there is $R \in (r, 1]_{\mathbb{R}}$ such that

$$f(z) = \|f_r\| \left(\frac{|z|}{r} \right)^{n'}, r < |z| \leq R. \quad (1.2)$$

where N is as in Equation (1.1), and $\|f\|_r = \sup\{|f(z)| \mid |z| \leq r\}$ where $r = 1$, Equation (1.2) has no analog as no R extends as $f(z)$ is undefined where $|z| > 1$. This means that N_1 is extended

Definition 1.11.

$$\text{Spa}(\mathbb{T}_m, \mathbb{T}_m^0) = \{\text{continuous valuations } \gamma \text{ of } \mathbb{T}_m \text{ s.t. } \forall f \in \mathbb{T}_m^\circ, \gamma(f) \leq 1\}.$$

Remark 1.5. $N_1(T) = (1, 1)$, $N_1(T^{2026}) = (1, 2026)$.

1.3 Spectral Spaces

Remark 1.6. $\emptyset$ is not irreducible.

Proposition 1.12. Let X be a topological space. For a point $x \in X$,

$$\overline{\{x\}} = \{y \in X \mid \text{every neighbourhood of } y \text{ in } X \text{ contains } x\}. \quad (1.3)$$

Definition 1.12. A point $x \in X$ is a generic point if $\overline{\{x\}} = X$, (i.e. $\{x\}$ is dense in X).

Notation 1.3. For a topological space X , we denote the collection of all open sets to be $\mathcal{O}_X$.

Definition 1.13. Let X be a topological space and $x, y \in X$. We say x, y are topologically distinguishable if there is $U \in \mathcal{O}_X$ such that either $x \in U$ and $y \notin U$ or $x \notin U$ and $y \in U$. They are otherwise indistinguishable.

Definition 1.14. A topological space X is T_0 or Kolmogorov if for any pair of points are distinguishable.

Definition 1.15. A topological space X is sober if every irreducible neighbourhood has precisely one generic point or equivalently if X is T_0 and every irreducible subsets has at least one generic point.

Remark 1.7. It is easy to see that these are local conditions.

Definition 1.16. A topological space is T_1 if for $x, y \in X$, $x \neq y$ implies there is $U \in \mathcal{O}_X$ such that $x \in U$ and $y \notin U$. Equivalently, every point is closed. It is T_2 or Hausdorff if we can take $U, V \in \mathcal{O}_X$ such that $U \cap V = \emptyset$ and $x \in U$ and $y \in V$. Or equivalently, the diagonal is closed in $X \times X$.

Definition 1.17. X is quasi-compact if every open covering admits a finite subcovering. A subset M of X is quasi-compact if it is a quasi-compact as a topological space with the subset topology from X .

Proposition 1.13. If X can be covered by finitely many quasi-compact open neighbourhoods then it is quasi-compact.

Proposition 1.14. Every closed subset of a quasi-compact space is again quasi-compact.

Proposition 1.15. Let $X \xrightarrow{f} Y$ be a continuous with X is quasi-compact and Y is Hausdorff, then f is closed (i.e. every closed set A of X has a closed image $f(A)$ in Y).

Definition 1.18. A subset $\mathcal{B} \subseteq \mathcal{O}_X$ is a topology basis for X if for any $U \in \mathcal{O}_X$, we have $U = \bigcup_{V \in \mathcal{B}_U} V$ where $\mathcal{B}_U = \{V \in \mathcal{B} \mid V \subseteq U\}$.

Remark 1.8. The subsets of the form

$$D(f) = \{\mathfrak{p} \in \text{Spec } A \mid f \notin \mathfrak{p}\}$$

form a topology basis for $\text{Spec } A$.

Proposition 1.16. For a topological space X , the following conditions are equivalent.

- i). $\underline{\mathcal{O}}_C(X) := \{\Omega \in \mathcal{O}_X \mid \Omega \text{ is quasi-compact}\}$ is a topology basis closed under finite intersection (including the empty intersection $\bigcap_{\emptyset} = X$).
- ii). There is a topology base $\mathcal{B} \subseteq \underline{\mathcal{O}}_C(X)$ such that $\mathcal{B}$ consists of quasi-compact and it is closed under the finite intersection in X .

Definition 1.19. A topological space X is spectral if it is sober and satisfies one of the conditions in Proposition 1.16.

Definition 1.20. A spectral map is a map $X \xrightarrow{f} Y$ between spectral spaces such that for any quasi-compact subset Ω in Y , $f^{-1}(\Omega)$ is quasi-compact subset of X .

Remark 1.9. Note that a spectral map is continuous.

Proposition 1.17. If $\mathcal{B} \subseteq \mathcal{O}_Y$ is a topology base as in the second condition of Proposition 1.16 for Y , then f is spectral if and only if $\forall \Omega \in \mathcal{B}, f^{-1}(\Omega) \in \underline{\mathcal{O}}_C(X)$.

Example 1.3. $\text{Spec } A$ is spectral and every ring homomorphism defines a spectral map and $\{D(f) \mid f \in A\}$ are such base as in Proposition 1.16 for $\text{Spec } A$.

Example 1.4. The underlying space of qcqs (i.e. quasi-compact and quasi-separated) scheme is spectral.

Remark 1.10. Every closed and every quasi-compact open sets of a spectral space is spectral.

Remark 1.11. A Noetherian space X (i.e. $\underline{\mathcal{O}}_C(X) = \mathcal{O}_X$) is spectral if and only if it is sober.

Proposition 1.18. Every non-empty closed subset of a spectral space X contains a closed point.

Proof. By Remark 1.10, it suffices to show the case for every spectral space $X \neq \emptyset$. Set,

$$\mathcal{M} = \{Z \subseteq X \mid Z \text{ is closed}\}.$$

Equip $\mathcal{M}$ with a partial order by $\subseteq$. Since X is quasi-compact, $\mathcal{M}$ has a minimal element with such order. Let Z be a such minimal element in $\mathcal{M}$. If Z has two different element x, y , by T_0 condition, we have $U \in \mathcal{O}_X$ such that $x \notin U$ and $y \in U$. Then $Z' = Z \setminus U$ is in $\mathcal{M}$. As $x \in Z'$, this contradicts to the minimality of Z as $y \in Z \setminus Z'$. $\square$

Definition 1.21. A filter base of a set M is a non-empty subset $\mathcal{B}$ of $\mathfrak{P}(M)$ such that

- i). $\emptyset \notin \mathcal{B}$,
- ii). for $N, O \in \mathcal{B}$ there is $P \in \mathcal{B}$ with $P \subseteq N \cap O$.

A filter is a filter base $\mathcal{F}$ such that for $N \in \mathcal{F}$, and $N \subseteq O \subseteq M$ implies $O \in \mathcal{F}$.

A filter generated by a filter base $\mathcal{B}$ is such that

$$\mathcal{F} := \{N \mid N \subseteq M \text{ and there is } O \in \mathcal{B} \text{ such that } O \subseteq N\}.$$

A direct image under a map $M \xrightarrow{f} X$ of a filter $\mathcal{F}$ is such that

$$f_*\mathcal{F} := \{Y \mid Y \subseteq X \text{ and } f^{-1}(Y) \in \mathcal{F}\}.$$

Example 1.5. Neighbourhood filter $\mathcal{N}_x$ of $x \in X$ is such that

$$\mathcal{N}_x := \{N \subseteq X \mid \exists U \in \mathcal{O}_X \text{ s.t. } x \in U \subseteq N\}.$$

Example 1.6. A principal filter $\mathcal{P}_x$ of $x \in X$ is such that

$$\mathcal{P}_x = \{N \subseteq X \mid x \in N\}.$$

Definition 1.22. A neighbourhood base for $x \in X$ is a filter base $\mathcal{B}$ such that it generates the neighbourhood filter of x .

Remark 1.12. The notion of filters is introduced by Cartan.

Proposition 1.19. For a filter $\mathcal{F}$ of X , the following statements are equivalent.

- 1). $\mathcal{F}$ is a $\subseteq$ -maximal element of the set of filters on X .
- 2). If $M, N \subseteq X$ and $M \cup N \in \mathcal{F}$ then $M \in \mathcal{F}$ or $N \in \mathcal{F}$.
- 3). For $M \subseteq X$, precisely one of M and $X \setminus M$ is in $\mathcal{F}$.

Definition 1.23. Such filter $\mathcal{F}$ satisfying the conditions in Proposition 1.19 is called an ultrafilter.

Theorem 1.1 (Tarski). Every filter on X is contained in some ultrafilter.

Proof. Follows from Zorn's lemma. □

Proposition 1.20. Let $\mathcal{U}$ be an ultrafilter of X then all direct images of $\mathcal{U}$ are ultrafilters. If $\iota : A \rightarrow X$ is the inclusion of a subset then this induces,

$$\{\text{ultrafilters on } A\} \xrightarrow{\iota} \{\text{ultrafilters on } X \text{ containing } A\},$$

which is a bijection.

Definition 1.24.

$$\text{Lim } \mathcal{F} = \{x \in X \mid \mathcal{N}_x \subseteq \mathcal{F}\}.$$

is the set of limits of $\mathcal{F}$. If it contains precisely one $x \in X$, we denote it by

$$\text{Lim } \mathcal{F} = x.$$

Definition 1.25. We say that $x \in X$ is a point of accumulation (denoted by *poa*, sometimes called *point of condensation*) if for a neighbourhood $V \in \mathcal{N}_x$ and $M \in \mathcal{F}$, we have $V \cap M \neq \emptyset$.

Proposition 1.21. Let $x \in X$ then

$$\overline{\{x\}} = \lim \mathcal{P}_x = \text{poa of the principal filter of } x = \text{Lim } \mathcal{N}_x.$$

Proposition 1.22. Every limit of a filter $\mathcal{F}$ is a poa of $\mathcal{F}$. If $\mathcal{F}$ is a subset of some filter $\mathcal{G}$, then every limit of $\mathcal{F}$ is a limit of $\mathcal{G}$ and every poa of $\mathcal{G}$ is a poa of $\mathcal{F}$. Furthermore if $\mathcal{U}$ is an ultrafilter, every poa of $\mathcal{U}$ is a limit of $\mathcal{U}$. For every filter, the set of its limit and of its points of accumulation points are closed. If $A \xrightarrow{\iota} X$ is an with induces topology then $\text{Lim } \mathcal{F} = A \cap \text{Lim } \iota_* \mathcal{F}$.

Proof. For the third assertion, if a is a poa of $\mathcal{U}$ and $V \in \mathcal{N}_x$, then $X \setminus V \notin \mathcal{U}$ as x is a poa. Hence $V \in \mathcal{U}$ by Proposition 1.19. For the forth assertion, let $x \notin \text{Lim } \mathcal{F}$, (respectively x not a poa of $\mathcal{F}$) then there is $U \in \mathcal{N}_x$ such that $U \notin \mathcal{F}$ (respectively, such that there is $M \in \mathcal{F}$ such that $U \cap M = \emptyset$). Then no element of U is a limit (respectively, poa) of $\mathcal{F}$. $\square$

Proposition 1.23. For a subset M of a topological space X , the following subsets all coincide.

- a). $\{\text{limits of ultrafilters } \mathcal{U} \text{ of } X \text{ such that } M \in \mathcal{U}\},$
- b). $\bigcup_{\mathcal{F} \text{ filter, } M \in \mathcal{F}} \text{Lim } \mathcal{F}.$
- c). $\{x \in X \mid \text{there is a filter } \mathcal{F} \text{ such that } M \in \mathcal{F} \text{ and } x \text{ is a poa of } \mathcal{F}\}.$
- d). $\bigcap_{M \subseteq Z \subseteq X, \text{closed}} Z.$

Proof. Denote those sets by M_a, M_b, M_c, M_d . Then,

$$M_a \underbrace{\subseteq}_{\text{trivial}} M_b \underbrace{\subseteq}_{\text{Proposition 1.22}} M_c \underbrace{\subseteq}_{\text{definition}} M_d.$$

For the reverse direction, $M_d = \bigcup_{U \in \mathcal{O}_X, U \cap M = \emptyset} (X \setminus U)$. Finally if $x \in M_d$, then

$$\{U \cap M \mid U \in \mathcal{N}_x\}$$

is a filter base containing M , by Theorem 1.1, it is contained in some ultrafilter $\mathcal{U}$ and $x \in \text{Lim } \mathcal{U}$. $\square$

Definition 1.26. The set of above is called the closure of M in X .

Remark 1.13. The closure of a point is a special case of this.

Corollary 1.1. For a subset A of a topological space X , the following conditions are equivalent.

- i). If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a limit of $\mathcal{F}$ then $x \in A$.
- ii). If $\mathcal{F}$ is a filter on X such that $A \in \mathcal{F}$ and x is a poa of $\mathcal{F}$ then $x \in A$.
- iii). If $\mathcal{U}$ is an ultrafilter on X such that $A \in \mathcal{U}$, every limit of $\mathcal{U}$ in X is in A .
- iv). A is closed.

Proof. Clear from Proposition 1.23 □

Proposition 1.24. For a map $f : X \rightarrow Y$ of topological spaces. The following conditions are equivalent.

- i). f is continuous.
- ii). If $\mathcal{F}$ is a filter on X and $x \in \text{Lim } \mathcal{F}$ then $f(x) \in \text{Lim } f_*\mathcal{F}$.
- iii). If $\mathcal{F}$ is a filter on X and $x \in X$ is a poa of $\mathcal{F}$, then $f(x)$ is a poa of $f_*\mathcal{F}$.
- iv). If $\mathcal{U}$ is an ultrafilter on X and $x \in \text{Lim } \mathcal{U}$, then $f(x) \in \text{Lim } f_*\mathcal{U}$.

Proof. i) $\Rightarrow$ ii) and iii) $\Rightarrow$ iv) are equivalent, essentially from Definition 1.25. iii) $\Rightarrow$ iv) trivial and due to Proposition 1.22. i) $\Rightarrow$ iv) is by Corollary 1.1 as $f^{-1}(A)$ is closed for any closed A in Y . □

Remark 1.14. By Definition 1.25, we have the following. The set of poa of a filter $\mathcal{F}$ is the intersection of the closure of the elements of $\mathcal{F}$ or the intersection of all $A \in \mathcal{F}$ which are closed in X .

Proposition 1.25. Let X be a topological space. The following conditions are equivalent.

- i). X is Hausdorff.
- ii). Every filter has at most one limit.
- iii). Every ultrafilter on X has at most one limit.

The following conditions are also equivalent.

- i'). X is quasi-compact.

Proof.

i) $\Rightarrow$ ii). If $x \neq y$, has disjoint neighborhoods $x \in U$ and $y \in V$. If $U \in \mathcal{F}$ and $V \in \mathcal{F}$, then $U \cap V \in \mathcal{F}$, which is not possible. Thus precisely one of x and y is in $\text{Lim } \mathcal{F}$.

ii) $\Rightarrow$ iii) is trivial.

iii) $\Rightarrow$ i). Assume $x \neq y$ and consider the set

$$\{U \cap V \mid U \in \mathcal{N}_x, V \in \mathcal{N}_y\}.$$

This is a filter □

Proposition 1.26. *Let X be a topological space. The following conditions are equivalent.*

i'). X is quasi-compact.

ii'). Every filter has at least one poa.

iii'). Every ultrafilter has at least one limit.

Proof.

i') $\Rightarrow$ ii') By 1.14,

$$\{x \mid x \text{ is a poa of } \mathcal{F}\} = \bigcap_{M \in \mathcal{F}} \overline{M} = \emptyset. \quad (1.4)$$

by quasi-compactness X , as every finite intersection is in $\mathcal{F}$ hence this is not $\emptyset$.

ii') $\Rightarrow$ iii'). By Proposition 1.22.

iii') $\Rightarrow$ i'). Let $(A_\lambda)_\Lambda$ be a family of closed subsets such that for any finite subset $F \subseteq \Lambda$, we have,

$$\bigcap_{\lambda \in F} A_\lambda \neq \emptyset.$$

Then the collection,

$$\left\{ \bigcap_{\lambda \in F} A_\lambda \mid F \subseteq \Lambda, |F| < \infty \right\}$$

is a filter base, thus it is contained in some ultrafilter $\mathcal{U}$. Let $x \in \text{Lim } \mathcal{U}$. For $\lambda \in \Lambda$, $A_\lambda \in \mathcal{U}$ hence $x \in A_\lambda$ by Corollary 1.1. Hence $x \in \bigcap_{\lambda \in \Lambda} A_\lambda$ and this intersection is not empty. $\square$

Proposition 1.27. *If X satisfies these equivalent conditions of Proposition 1.16, then for every ultrafilter $\mathcal{U}$ on X , the closed subset $\text{Lim } \mathcal{U} \subseteq X$ is irreducible.*

Proof. Let $\Omega, \Theta \in \underline{\Omega}_C(X)$ such that $\Omega \cap \text{Lim } \mathcal{U} \neq \emptyset$ and $\Theta \cap \text{Lim } \mathcal{U} \neq \emptyset$. By the definition of $\text{Lim } \mathcal{U}$, $\Omega, \Theta \in \mathcal{U}$. Hence $\Omega \cap \Theta \in \mathcal{U}$. As $\Omega \cap \Theta$ is quasicompact $\mathcal{U} \cap \mathcal{O}_{\Omega \cap \Theta}$ has a limit $x \in \Omega \cap \Theta$ by Proposition 1.25. By Proposition 1.22, $x \in \text{Lim } \mathcal{U}$.

If U and V are open subsets such that $U \cap \text{Lim } \mathcal{U}$ and $V \cap \text{Lim } \mathcal{U}$ are not emptyset, there are $\Omega \subseteq U$ and $\Theta \subseteq V$ as above (as $\underline{\Omega}_C(X)$ is a topology base), hence $U \cap V \cap \text{Lim } \mathcal{U}$ is not empty. Finally by Proposition 1.25, $\text{Lim } \mathcal{U} \neq \emptyset$ as X is quasi-compact. $\square$

Definition 1.27. *We say that $x \in X$ is a generic limit of a ultrafilter $\mathcal{U}$ if $\text{Lim } \mathcal{U} = \{x\}$.*

Definition 1.28. *Let X be a topological space in which every ultrafilter $\mathcal{U}$ has a unique generic limit. Then we write,*

$$\text{glim } \mathcal{U} = \text{glim}_X \mathcal{U}$$

We say that a subset $M \subseteq X$ is closed under generic limit in X if $M \in \mathcal{U}$ for every ultrafilter $\mathcal{U}$ implies $\text{glim } \mathcal{U} \in M$.

Proposition 1.28.

- 1). X is T_0 if and only if every ultrafilter on X has at most one generic limit.
- 2). If every ultrafilter on X has some generic points then every closed subset $Z \subseteq X$ has some generic points.

Proof. For 1). the sufficiency of being T_0 is by Definition 1.15. If x and y are topologically indistinguishable, then $\overline{\{x\}} = \overline{\{y\}} = Z$ for some Z . Furthermore, the principal filter $\mathcal{U}$ of x which has $\text{Lim } \mathcal{U} = Z$ from Example 1.6 has both x and y as generic limits.

2). Irreducibility of Z implies that $\mathcal{O}_Z \setminus \{\emptyset\}$ is a filter base in X . Hence it is contained in some ultrafilter $\mathcal{U}$. By Definition 1.26 and the closedness of Z implies $Z = \text{glim } \mathcal{U}$. Hence every $\text{glim } \mathcal{U}$ is a generic point of Z . $\square$

Proposition 1.29. Assume every ultrafilter $\mathcal{U}$ on X has a unique generic limit $\text{glim } \mathcal{U}$. Then the following statements hold.

- 1). The set of subsets of X which are closed under glim is closed under arbitrary intersection and finite unions.
- 2). An open subset $V \subseteq X$ is closed under glim if and only if V is quasi-compact.

Proof. For 1), the closedness under the intersections is obvious. If $M \cup N \in \mathcal{U}$ then $M \in \mathcal{U}$ or $N \in \mathcal{U}$ by Proposition 1.19. Hence $\text{glim } \mathcal{U} \in M \subseteq M \cup N$ or $\text{glim } \mathcal{U} \in N \subseteq M \cup N$.

Let $\mathcal{U}$ be an ultrafilter on V and $\mathcal{V}$ be the image of $\mathcal{U}$ under the embedding $V \hookrightarrow X$. If V is closed under glim then $\text{glim } \mathcal{V} \in V$ is a limit of $\mathcal{U}$ by Proposition 1.22. By Proposition 1.25, V is quasi-compact. If V is quasi-compact, $\mathcal{U}$ has a limit $x \in V$ and $x \in \text{Lim } \mathcal{V}$ by Proposition 1.22. Hence $x \in \overline{\{y\}}$ hence $y \in V$ and V is closed. $\square$

Definition 1.29. A spectral space X is quasi-separated if for any ultrafilter $\mathcal{U}$ on X , if its limit $\text{Lim } \mathcal{U}$ is non-empty then it is an irreducible closed subset.

Proposition 1.30.

- 1). A topological space X is separated if and only if every ultrafilter on X has a unique generic limit and the set of those $\Omega \in \mathcal{O}_X$ which are closed under glim is a topology base.
- 2). An open subset of a separated space X is quasi-compact if and only if it is closed under glim . (This is proved in Proposition 1.29).
- 3). A map $f : X \rightarrow Y$ is separated if and only if it is continuous and $\text{glim}_Y f_* \mathcal{U} = f(\text{glim}_X \mathcal{U})$ for every ultrafilter $\mathcal{U}$ on X . (This is proved by the above assertion).

Proof. If X is separated, then by Proposition 1.27, $\text{Lim } \mathcal{U}$ is always closed and irreducible subset, hence $\text{glim } \mathcal{U}$ exists uniquely. By Proposition 1.29, one has $\{U \in \mathcal{O}_X \mid U \text{ is closed under glim}\} = \underline{\Omega}_C(X)$ is a topology base. If any ultrafilter has a unique generic limit, then X is sober by Proposition 1.29 and $\mathcal{O}_C(X) = \{U \in \mathcal{O}_X \mid U \text{ is closed under glim}\}$ which is closed under finite intersection, in X and by the assumption this is a topology base for X . $\square$

Theorem 1.2. *A topological space X is spectral if and only if there is a ring A such that X is homeomorphic to $\text{Spec } A$.*

Proof. We omit the hard part which is $\Rightarrow$ direction (the existence of such ring A for every spectral space X). That $\text{Spec } A$ is spectral can be seen as follows. Let $\mathcal{U}$ be an ultrafilter on $\text{Spec } A$ and

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid V(a) \in \mathcal{U}\} = \{a \in A \mid \{\mathfrak{p} \in \text{Spec } A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

One $\mathfrak{g}_{\mathcal{U}} \in \text{Spec } A$ that is let $a_1, a_2 \in \mathfrak{g}_{\mathcal{U}}$, and $M_i = \{\mathfrak{p} \in \text{Spec } A \mid a_i \in \mathfrak{p}\} \in \mathcal{U}$ hence $M_1 \cap M_2 \in \mathcal{U}$, hence $\{\mathfrak{p} \mid a_1 + a_2 \in \mathfrak{p}\} \in \mathcal{U}$. as this contains $M_1 \cap M_2$. Hence $a_1 + a_2 \in \mathfrak{g}_{\mathcal{U}}$ thus it is an ideal. Let $a_1 a_2 \in \mathfrak{g}_{\mathcal{U}}$, but $a_2 \notin \mathfrak{g}_{\mathcal{U}}$. Then $M_2 \notin \mathcal{U}$ but $M_1 \cup M_2 \in \mathcal{U}$ since $a_1 a_2 \in \mathfrak{p}$, one of them must belong to it. Hence

$$M_1 \cup M_2 \supseteq \{\mathfrak{p} \mid a_1 a_2 \in \mathfrak{p}\} \in \mathcal{U}.$$

But $M_1 \cup M_2 \in \mathcal{U}$ and $M_2 \notin \mathcal{U}$ implies $M_1 \in \mathcal{U}$ hence $a_1 \in \mathfrak{g}_{\mathcal{U}}$. It is also clear that $\mathfrak{g}_{\mathcal{U}} \in D(f)$ if and only if $D(f) \in \mathcal{U}$. By Lemma 1.1, $\mathfrak{g}_{\mathcal{U}}$ is $\text{glim } \mathcal{U}$ and all $D(f)$ are glim -closed. $\square$

Lemma 1.1. *Let $\mathcal{B}$ be a topology base of the topological space X .*

- a) *Let $\mathcal{U}$ be an ultrafilter on X and $g \in X$ such that for all $\Omega \in \mathcal{B}$, $g \in \Omega \Leftrightarrow \Omega \in \mathcal{U}$. Then g is a generic limit of $\mathcal{U}$.*
- b) *If X is T_0 and for every ultrafilter $\mathcal{U}$ on X , there is $g \in X$ as in a), then X is spectral and the elements of $\mathcal{B}$ are quasi-compact.*

Proof. Let $U \in \mathcal{N}_x$. Then there is $\Omega \in \mathcal{B}$ such that $x \in \Omega \subseteq U$. By our assumption, $\Omega \in \mathcal{U}$ hence $U \in \mathcal{U}$ thus g is a limit of $\mathcal{U}$. Let $x \notin \overline{\{g\}}$ then there is $U \in \mathcal{N}_x$ such that $g \notin U$. As $\mathcal{B}$ is a topology base there is $\Omega \in \mathcal{B}$ such that $x \in \Omega \subseteq U$. Then $g \notin \Omega$. Hence $\Omega \in \mathcal{U}$. Since Ω is a neighbourhood of x , this shows $x \notin \text{Lim } \mathcal{U}$. Thus g is a generic limit. This proves a).

For b), it follows from our assumption and a) that X has unique generic limits of ultrafilters and the elements of $\mathcal{B}$ are closed under such generic limits. The claim now follows from Proposition 1.30. $\square$

Remark 1.15. *If $\mathcal{U}$ is an ultrafilter on $\text{Spec } A$ and*

$$\mathfrak{g}_{\mathcal{U}} = \{a \in A \mid \{\mathfrak{p} \in \text{Spec } A \mid a \in \mathfrak{p}\} \in \mathcal{U}\}.$$

Then the lemma applies with $\mathcal{B} = \{D(a) \mid a \in A\}$.

Corollary 1.2 (Of Lemma 1.1 b), and the proof of Theorem 1.2). *The subset $D(a) \subseteq \text{Spec } A, a \in A$ are quasi-compact.*

Proof. From Remark 1.15 and Lemma 1.1. $\square$

Definition 1.30. *Let X be a topological space. A subset U which is open and closed is called clopen.*

Corollary 1.3 (Of Proposition 1.30). *For a compact (thus Hausdorff) X , the following conditions are equivalent.*

a). *X is spectral.*

b). *The clopen subsets of X form a topology base.*

In this case, the open subsets U is quasi-compact if and only if it is clopen and when Y is a spectral space a map $Y \rightarrow X$ is spectral it is continuous.

Proof. X has unique generic limits which are the generic limits of ultrafilters and by Corollary 1.1, the open subsets which are invariant under these limits are preimages of the open subsets are also closed.

To show these two are equivalent. If U is closed, it is quasi-compact as it is a closed subset of a quasi-compact space. If U is quasi-compact then it is closed by Proposition 1.15. If $Y \xrightarrow{f} X$ is continuous and $\Omega \in \underline{Q}_C(X)$, then Ω is clopen hence $f^{-1}\Omega$ is clopen in Y , hence $f^{-1}\Omega \in \underline{Q}_C(Y)$. $\square$

Definition 1.31. *A set $\mathcal{A}$ of subsets of X is a boolean algebra if the following conditions hold.*

1. $\emptyset \in \mathcal{A}$.
2. $M \in \mathcal{A} \Rightarrow X \setminus M \in \mathcal{A}$.
3. $\mathcal{A}$ is closed under taking finite unions.

Remark 1.16. *From the basis axioms of boolean algebras. Clearly we have $X \in \mathcal{A}$ and $M_1, M_2 \in \mathcal{A} \Rightarrow M_1 \cap M_2 \in \mathcal{A}$.*

Proposition 1.31. *Let X be a spectral space. The following boolean algebras of subsets of X coincide.*

1. *The boolean algebra generated by $\underline{Q}_C(X)$.*
2. *$\{M \subseteq X \mid M \text{ is closed under generic limits of ultrafilters}\}$.*

Let $\mathcal{K}_X$ be this boolean algebra of subsets of X . The topology on X obtained by taking $\mathcal{K}_X$ as a topology base coincides with the one for which preserving the subsets $M \subseteq X$ which are closed under glim of ultrafilters are closed subsets.

Finally Let X_{con} be X equipped with this topology. Then X_{con} is a spectral hausdorff space (i.e. X_{con} as in Corollary 1.2), and $X_{\text{con}} \xrightarrow{\text{id}_X} X$ is a spectral map.

For every ultrafilter $\mathcal{U}$ of X , $\lim_{X_{\text{con}}} \mathcal{U} = \text{glim}_X \mathcal{U}$.

Not Required for the Exam. Let $\mathcal{K}_1$ be the Boolean algebra generated by $\underline{\mathcal{Q}}_C(X)$ and

$$\mathcal{K}_2 := \{M \mid M \text{ and } X \setminus M \text{ closed under glim}\}.$$

By Proposition 1.28, $\mathcal{K}_2$ is a Boolean algebra. By Proposition 1.30 b), and Corollary 1.1, $\underline{\mathcal{Q}}_C(X) \subseteq \mathcal{K}_2$, hence we have an inclusion $\mathcal{K}_1 \subseteq \mathcal{K}_2$. For the time being, $\mathcal{K}_x := \mathcal{K}_1$. Let X_1 be the topological space with underlying space X and the topology generated by $\mathcal{K}_x$. Let X_2 be the topological space again with the same underlying set such that for a subset $A \subseteq X$ is closed if and only if it is closed under glim of ultrafilters. By Proposition 1.28, X_2 is indeed a topological space. Since $\mathcal{K}_x \subseteq \mathcal{K}_2$, X_2 is finer or equal to X_1 .

The space X_1 is Hausdorff. If $x \neq y$ then without loss of generality exchanging x and y if necessary, there is $U \in \mathcal{O}_X$ such that $x \in U$ and $y \notin U$. As $\underline{\mathcal{Q}}_C(X)$ is a topology base, hence without loss of $U \in \underline{\mathcal{Q}}_C(X)$. Then $U \in \mathcal{K}_X$ and $X \setminus U \in \mathcal{K}_X$ and they are distinguished neighbourhoods of x and y in X_1 . Moreover, X_2 is quasi-compact by Proposition 1.25. Thus it suffices to consider an ultrafilter $\mathcal{U}$ on X and to show that it has a limit in X_2 . Let $g = \text{glim}_X \mathcal{U}$. We claim that $g \in \text{Lim}_{X_2} \mathcal{U}$. For this, let $U \in \mathcal{O}_{X_2}$ such that $g \in U$, we must show $U \in \mathcal{U}$. Otherwise $X \setminus U \in \mathcal{U}$. But by the definition of X_2 , $X \setminus U$ is closed under glim of ultrafilters. Thus $g \in X \setminus U$ which is a contradiction.

By Proposition 1.15, the continuous map $X_2 \xrightarrow{\text{id}_X} X$ is a homeomorphism. Thus first two parts are shown.

Let $X_{\text{con}} = X_1 = X_2$. Let $\Omega \in \mathcal{K}_2$. By Corollary 1.1, Ω is clopen in X_{con} . Since $\mathcal{K}_1$ is a topology base, Ω is a union of elements of $\mathcal{K}_1$ and since $\Omega \in \underline{\mathcal{Q}}_C(X_{\text{con}})$, the union can be chosen to be finite. Then $\Omega \in \mathcal{K}_1$ as $\mathcal{K}_1$ is a Boolean algebra of subsets.

From now on, we will regard $\mathcal{K}_x = \mathcal{K}_1 = \mathcal{K}_2$. Since the elements of $\mathcal{K}_x$ are clopen and $\mathcal{K}_x$ is a topology base for X_{con} , and X_{con} compact. By Corollary 1.3, X_{con} is spectral. By Proposition 1.30, $X_{\text{con}} \xrightarrow{\text{id}_X} X$ is spectral. $\square$

Definition 1.32. *The elements of $\mathcal{K}_x$ are called constructible subsets and the topology of X_{con} the constructible topology.*

Corollary 1.4. *A subset of X is an element of $\mathcal{K}_x$ if and only if it is clopen in X_{con} .*

Corollary 1.5 (Of Proposition 1.30). *A map $X \xrightarrow{f} Y$ between spectral spaces is continuous if and only if spectral is continuous and $X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ is continuous.*

Corollary 1.6. *For a spectral space X , $\underline{\mathcal{Q}}_C(X) = \mathcal{O}_X \cap \mathcal{K}_X$.*

Definition 1.33. *Let X be a spectral. We say that $x \in X$ is a specialization of $y \in X$ and conversely y is a generalization of x if $x \in \overline{\{y\}}$. We denote such relation by $x \preceq y$.*

Proposition 1.32.

1. *A subset $Z \subseteq X$ is closed in X if and only if it is closed in X_{con} and closed under specialization. That is $x \preceq y, y \in Z \Rightarrow x \in Z$.*

2. A subset $U \subseteq X$ is open in X if and only if it is open in X_{con} and closed under generalization.
3. Let Z be closed in X_{con} and $\overline{Z}$ be its closure in X . Then $\overline{Z}$ is the set of specialization of elements of Z .

Proof. If $z \in Z$ and $x \preceq z$, then $x \in \overline{\{z\}} \subseteq \overline{Z}$, there are ???. If $x \in \overline{Z}$ by Proposition 1.23, there is an ultrafilter $\mathcal{U}$ on X such that $Z \in \mathcal{U}$ and $x \in \text{Lim} \mathcal{U}$. Since Z is closed in X_{con} , $g = \text{glim} \mathcal{U} \in Z$. But $x \preceq g$, proving the third statement. The first two are equivalent and the first follows from the third. $\square$

Corollary 1.7. *For a spectral space X , the following statements are equivalent.*

1. X is T_0 .
2. $X = X_{\text{con}}$ as topological space.
3. Every quasi-compact subset of X is closed.
4. X is Hausdorff.
5. X is T_4 .

Corollary 1.8. *For a map $X \xrightarrow{f} Y$ between spectral spaces, the following statements are equivalent.*

1. $\text{glim}_Y f_* \mathcal{U} = f(\text{glim}_X \mathcal{U})$ for all ultrafilters $\mathcal{U}$ on X , and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.
2. $X_{\text{con}} \xrightarrow{f} Y_{\text{con}}$ and $x \preceq_X \xi$ implies $f(x) \preceq_Y f(\xi)$.
3. f is spectral.

Corollary 1.9. *For a subset $Z \subseteq X$, the following statements are equivalent.*

- a). Z is spectral with induced topology and $Z \rightarrow X$ is spectral.
- b). Z is closed in X_{con} .

In this case, we denote Z_{con} to be the topology on Z induced from topology X_{con}

1.4 The Valuation Spectrum of a Rings

Definition 1.34. *An ordered abelian group is an abelian group Γ (under multiplication) with a linear ordering $\leq$ such that*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

Example 1.7. $\{1\}$ and $(0, \infty)_{\mathbb{R}}^{\times}$ are ordered abelian groups.

Remark 1.17. *The inequality,*

$$\gamma_1 \leq \gamma_2, g_1 \leq g_2 \Rightarrow \gamma_1 g_1 \leq \gamma_2 g_2.$$

is strict if one of the inequalities in the left hand side is strict. In particular, $g > 1$ implies $g^k > 1$ and $g < 1$ implies $g^k < 1$ for all $k \in \mathbb{N}$. Thus Γ has no torsion.

Definition 1.35. *A (general) valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group.*

Definition 1.36. *A valuation of a ring R is a map $R \xrightarrow{\gamma} \Gamma \cup \{0\}$ where Γ is an ordered abelian group with multiplication such that the following hold.*

$$i). \forall x \in \Gamma \cup \{0\}, 0 \cdot x = x \cdot 0 = 0, 0 \leq x.$$

$$ii). \gamma(xy) = \gamma(x)\gamma(y)$$

$$iii). \gamma(1) = 1$$

$$iv). \gamma(0) = 0$$

$$v). \gamma(x + y) \leq \max\{\gamma(x), \gamma(y)\}.$$

Remark 1.18.

$$\gamma(-1) = 1, \quad \gamma(y) < \gamma(x) \Rightarrow \gamma(x + y) = \gamma(x).$$

Definition 1.37. *Let Γ_γ be the subgroup of Γ generated by $\{\gamma(r) \mid r \in R\} \setminus \{0\}$. We say that valuations γ and ω are equivalent if there is an order preserving isomorphism $\Gamma_\gamma \xrightarrow{i_\sim} \Gamma_\omega$ such that*

$$\omega(r) = i(\gamma(r)),$$

where we put $i(0) = 0$.

Proposition 1.33. *The valuation of r and w are equivalent if and only if*

$$\{(r, s) \in R^2 \mid \gamma(r) \leq \gamma(s)\} = \{(r, s) \in R^2 \mid w(r) \leq w(s)\}.$$

Definition 1.38.

$$\text{supp } \gamma = \{r \in R \mid \gamma(r) = 0\}.$$

Remark 1.19. *$\text{supp } \gamma$ is a prime ideal. Indeed, $\gamma(r) = 0$ then $\gamma(r, s) = \gamma(r)\gamma(s) = 0$ with $s \in R$, if $r, s \in \text{supp } \gamma$ then $\gamma(r + s) \leq \max\{\gamma(r), \gamma(s)\} = 0$. Clearly $1 \notin \text{supp } \gamma$. If $\gamma(rs) = 0$, then $\gamma(r)\gamma(s) = 0$. Thus $\gamma(r) = 0$ or $\gamma(s) = 0$.*

Remark 1.20. *The support only depends on the equivalence classes.*

Example 1.8. If $\mathfrak{p}$ is a prime ideal and

$$\gamma(r) = \begin{cases} 1, & r \notin \mathfrak{p}, \\ 0, & \text{quadr} \in \mathfrak{p}. \end{cases}$$

the trivial valuation belonging to $\mathfrak{p}$.

Definition 1.39. A valuation is trivial if it is equivalent to this kind of valuation. That is $\Gamma_\gamma = \{1\}$.

Example 1.9. If $R = K$ is a field, then $\gamma(s) = 1$ where $\gamma^k = 1$ for some $k \in \mathbb{N}$ as Γ has no torsion. Then for a finite field only admits trivial valuations.

Example 1.10. The p -adic valuation of $\mathbb{Q}$ or $\mathbb{Q}_p$.

Definition 1.40. Let $\text{Spr}(R)$ be the set of equivalent classes of valuation of R . For $x \in \text{Spr}(R)$, we choose a representative $|\cdot|_x$. Let Γ_x be the group generated by those $|r|_x$ which are non-zero. For $m \in \mathbb{N}$ and $(r_i)_{i=0}^m \in \mathbb{R}^{m+1}$, let

$$\mathcal{R}(r_0|r_1, \dots, r_m) = \mathcal{R}_{\text{Spr}(R)}(r_0|r_1, \dots, r_m) = \{x \in \text{Spr}(R) | 0 \neq |f_0|_x \geq |f_i|_x, 1 \leq i \leq m\}.$$

These are called rational open subsets. We equip $\text{Spr}(R)$ with the topology for which the set of rational open subsets is a topology base.

If $A \xrightarrow{\alpha} B$ is a ring homomorphism then this induces

$$\text{Spr}(B) \xrightarrow{\text{Spr}(\alpha)} \text{Spr}(A)$$

sends $x \in \text{Spr}(B)$ to the equivalence class of the valuation,

$$a \mapsto |\alpha(a)|_x$$

of A . We set,

$$\gamma(a) = \text{Spr}(A) \setminus \mathcal{R}(a|).$$

Remark 1.21. The topology there is the coordinated? for which

$$\text{Spr} \xrightarrow{x \mapsto \text{supp}(x)} \text{Spec}$$

is continuous and all

$$\{x \in \text{Spr}(A) \setminus \mathcal{R}(a) \mid |\alpha|_x \leq |a|_x\}$$

for $(a, \alpha) \in A^2$ are open.

Remark 1.22. The set of rational open subset is closed under finite intersections in $\text{Spr}(R)$. Indeed,

$$\mathcal{R}(1|) = \text{Spr}(R),$$

$$\mathcal{R}(r_0|r_1, \dots, r_m) \cap \mathcal{R}(s_0|s_1, \dots, s_n) = \mathcal{R}(r_0 s_0 | r_i s_j, 0 \leq i \leq n, 0 \leq j \leq n).$$

If you assume $m \leq 1$ or even $m = 1$, you only get a topology subbase.

One way of constructing generic limits of ultrafilters in $\text{Spr}(R)$ would be to use Proposition 1.33 and

$$\{(r, s) \in R^2 \mid |r|_g \leq |s|_g\} := \{(r, s) \in R^2 \mid \{x \in \text{Spr}(R) \mid |r|_x \leq |s|_x\} \in \mathcal{U}\}.$$

This would work and was shown by Huber.

Definition 1.41. Let $\mathcal{U}$ be an ultrafilter on Λ and $(X_\lambda)_{\lambda \in \Lambda}$ be a family of sets indexed by Λ . We call the elements $(x_\lambda)_{\lambda \in \Lambda}$ and $(\xi_\lambda)_\Lambda$ of $\prod_{\lambda \in \Lambda} X_\lambda$ $\mathcal{U}$ -equivalent if

$$\{\lambda \mid x_\lambda = \xi_\lambda\} \in \mathcal{U}.$$

The ultraproduct $\prod_{\mathcal{U}} X_\lambda$ is the set of $\mathcal{U}$ -equivalence classes of elements of $\prod_{\lambda \in \Lambda} X_\lambda$. If X_λ -s have group structures, $(x\xi) \in \prod_{\mathcal{U}} X_\lambda$ is given as the $\mathcal{U}$ -equivalence classes of $(x_\lambda \xi_\lambda)_\Lambda$. If X_λ -s are equipped with linear ordering, put

$$(x_\lambda)_{\lambda \in \Lambda} / \sim_{\mathcal{U}} \leq (\xi_\lambda)_{\lambda \in \Lambda} / \sim_{\mathcal{U}}$$

if $\{\lambda \in \Lambda \mid x_\lambda \leq \xi_\lambda\} \in \mathcal{U}$.

Proposition 1.34. Suppose $X_\lambda \neq \emptyset$ for all $\lambda \in \Lambda$.

1. If $\mathcal{U} = \{\Theta \subseteq \Lambda \mid \lambda_0 \in \Theta\}$ is the principal ultrafilter belonging to λ_0 , then $\prod_{\mathcal{U}} X_\lambda = X_{\lambda_0}$.
2. If F is a finite set and all $\lambda \in \Lambda, X_\lambda = F$. Then $\prod_{\mathcal{U}} F = F$ as $\Lambda = \bigcup_{f \in F} \{\lambda \mid x_\lambda = f\}$ (i.e. disjoint finite union hence there is $??$ are $f \in F$ such that $\{\lambda \mid x_\lambda = f\} \in \mathcal{U}$ where $(x)_\lambda = \prod_{\Lambda} F$)
3. If $X_\lambda = X'_\lambda \cup X''_\lambda$ (disjoint unions, $\forall \lambda \in \Lambda, X'_\lambda, X''_\lambda \neq \emptyset$) Then $\prod_{\mathcal{U}} (X'_\lambda \cup X''_\lambda) = (\prod_{\mathcal{U}} X'_\lambda) \cup (\prod_{\mathcal{U}} X''_\lambda)$ which is also a disjoint union as $\Lambda = \{\lambda \in \Lambda \mid x_\lambda \in X'_\lambda\} \cup \{\lambda \in \Lambda \mid x_\lambda \in X''_\lambda\}$.
4. Then $\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) = (\prod_{\mathcal{U}} \Gamma_\lambda) \cup \{0\}$ and $\prod_{\mathcal{U}} \Gamma_\lambda$ is an ordered abelian group where all Γ_λ are ordered abelian groups.

Remark 1.23. When some of the X_λ may be empty, it is better to define $\prod_{\mathcal{U}} X_\lambda$ as the set of equivalent classes of pairs (M, ξ) where $M \in \mathcal{U}$ is a subset and ξ a function defined on M such that $\xi(\lambda) \in X_\lambda$ where $\lambda \in M$. Two pairs $(M, \xi), (N, \nu)$ are equivalent if

$$\{\lambda \in M \cap N \mid \xi(\lambda) = \nu(\lambda)\} \in \mathcal{U}.$$

With this, we define,

$$\prod_{\mathcal{U}} \left(\bigcup_{i=1}^m X_\lambda^{(i)} \right) \cong \bigcup_{i=1}^m \prod_{\mathcal{U}} X_\lambda^{(i)}.$$

Remark 1.24. $\prod_{\mathcal{U}} (\text{ordered abelian group})$ is again an ordered abelian group and

$$\prod_{\mathcal{U}} (\Gamma_\lambda \cup \{0\}) \cong \left(\prod_{\mathcal{U}} \Gamma_\lambda \right) \cdot \{0\}.$$

Proposition 1.35. *For every ring R , $\text{Spr}(R)$ is a spectral space and a subset of the form $\mathcal{R}(f_0|f_1, \dots, f_m)$ with $m \in \mathbb{N}$ and $(f_i)_{i=0}^m \in R^{m+1}$ is quasi-compact.*

The generic limit of the ultrafilter $\mathcal{U}$ on $\text{Spr } R$ is given by the equivalence class of the valuation,

$$R \xrightarrow{r \mapsto (\text{the } \mathcal{U}\text{-equivalence class of } (|r|_\gamma)_{\gamma \in \text{Spr } R})} \left(\prod_{\gamma \in \text{Spr } R_{\mathcal{U}}} \Gamma_\gamma \right) \cup \{0\} = \prod_{\mathcal{U}} (\Gamma_\gamma \cup \{0\}). \quad (1)$$

Moreover, the map,

$$\text{Spr}(R) \xrightarrow{\gamma \mapsto \text{supp } |\cdot|_\gamma} \text{Spec } R$$

is spectral and for every ring homomorphism $R \xrightarrow{\rho} S$, the map

$$\text{Spr}(S) \xrightarrow{\text{Spr}(\rho)} \text{Spr}(R)$$

is also spectral.

Proof. It is obvious from the definition and the properties of ultrafilters that (1) is indeed a valuation. To show that it defines a generic limit of $\mathcal{U}$ and that the rational open subsets are quasi-compact, we use Lemma 1.1. Set

$$\mathcal{B} = \{\mathcal{R}(f_0|f_1, \dots, f_m) \mid m \in \mathbb{N}, (f_i)_{i=0}^m \in R^{m+1}\}.$$

if g ??? the equivalence class of (1),

$$[g \in \mathcal{R}(f_0|f_1, \dots, f_m)] \Leftrightarrow |f_0|_g \neq 0 \text{ and all } |f_1|_g \leq |f_0|_g,$$

$$\stackrel{\text{By (1)}}{\Leftrightarrow} (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}} \neq 0 \text{ and } (|f_i|_x)_{x \in X} / \sim_{\mathcal{U}} \leq (|f_0|_x)_{x \in X} / \sim_{\mathcal{U}},$$

$$\stackrel{\substack{\text{Definition of } \prod_{\mathcal{U}} \\ \text{and the valuation on it}}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \in \mathcal{U} \text{ and all } \{x \in X \mid |f_i(x)| \leq |f_0(x)|\} \in \mathcal{U},$$

$$\stackrel{\cap_{i=0}^m M_i \in \mathcal{U} \Leftrightarrow \forall M_i \in \mathcal{U}}{\Leftrightarrow} \{x \in X \mid |f_0|_x \neq 0\} \cap \bigcap_{i=1}^m \{x \in X \mid |f_i(x)| \leq |f_0(x)|\} \in \mathcal{U},$$

$$\stackrel{\text{Definition of rational open}}{\Leftrightarrow} \mathcal{R}(f_0 \mid f_1, \dots, f_m) \in \mathcal{U}.$$

Lemma 1.1 can indeed be applied. That the induced map $\text{Spr}(R) \xrightarrow{\text{supp}} \text{Spec}(R)$ is spectral follows from

$$\text{supp}(D(f)) = \mathcal{R}(f|).$$

Proposition ???. That $\text{Spr}(\rho)$ is spectral follows immediately from

$$\text{Spr}(\rho)^{-1} \mathcal{R}(f_0|f_1, \dots, f_m) = \mathcal{R}(\rho(f_0)|\rho(f_1), \dots, \rho(f_m)).$$

□

Remark 1.25.

Proposition 1.36. *For every ideal $I \subseteq R$, if $R \xrightarrow{\pi} R/I$ denotes the canonical projection, then the induced map $\text{Spr}(R/I) \xrightarrow{\pi} \text{Spr}(R)$ is a homeomorphism. Explicitly, we have,*

$$\text{Spr}(R/I) = V_{\text{Spr}(R)}(I) := \{x \in \text{Spr}(R) \mid \text{supp } x \supseteq I\} = \text{Spr}(R) \setminus \bigcup_{f \in I} \mathcal{R}(f). \quad (1.5)$$

If I is finitely generated (i.e. $I = \langle g_1, \dots, g_m \rangle_R$), then the right hand side of Equation (1.5) can be written as,

$$\text{Spr}(R/I) = \bigcup_{i=1}^m \mathcal{R}(g_i)$$

which is the embedded subset. Alternatively, we have,

$$V_{\text{Spr}(R)} = \{x \in \text{Spr}(R) \mid \forall f \in I, |f|_x = 0\}.$$

Furthermore, we can identify this with a homeomorphism,

$$\text{Spr}(R/I) \xrightarrow[\simeq]{\text{Spr}(\pi)} V(I).$$

Let S be a multiplicative subset of S and $\iota : R \rightarrow R_S$ be a canonical inclusion then,

$$\text{Spr}(R_S) = \xrightarrow[\text{Spr}(\iota)]{\simeq} \bigcap_{s \in S} (\text{Spr}(R) \setminus V(s)) = \bigcap_{s \in S} \mathcal{R}(s). \quad (1.6)$$

Proposition 1.37. *Let s be a specialization of $\eta \in \text{Spr}(R)$, then*

$$\text{supp } \eta \subseteq \text{supp } s. \quad (1.7)$$

Proof.

$$r \notin \text{supp } s \iff s \in \mathcal{R}(r) \Rightarrow \eta \in \mathcal{R}(r) \iff r \notin \text{supp } \eta. \quad (1.8)$$

□

Definition 1.42. *Let $\gamma \in \Gamma \cup \{0\}$ and $R \xrightarrow{v} \Gamma \cup \{0\}$ be a valuation. We then define,*

$$K_{v \leq \gamma} := \{r \in R \mid v(r) \leq \gamma\}.$$

For an equivalence class $x \in \text{Spr}(R)$ where $v \in x$, we may denote,

$$K_{x \leq \gamma} = K_{v \leq \gamma}.$$

For $\rho \in R$, we denote,

$$K_{v \leq \rho} := K_{v \leq v(\rho)}, K_{x \leq \rho} := K_{x \leq |\rho|_x}.$$

Similarly,

$$K_{v < \gamma} = K_{x < \gamma} := \{r \in R \mid v(r) < \gamma\}$$

$$K_{v < \rho} = K_{x < \rho} := K_{v < v(\rho)}.$$

Proposition 1.38. $K_{v \leq 1}$ is an integrally closed subring of R , and $K_{v < 1}$ is a prime ideal in $K_{v \leq 1}$.

Proof. These being a subring and an ideal can be checked easily. Let $r \notin K_{v \leq 1}$, then if m is a positive integer and $(p_i)_{i=1, \dots, m} \in K_{v \leq 1}^m$, we have,

$$\begin{aligned} v(r^m) &= v(r)^m, \\ &> v(r)^{m-1} \leq \max_{1 \leq i \leq m-1} v(p_i) v(r)^i, \\ &= \max_{1 \leq i \leq m-1} v(p_i r^i), \\ &\leq v \left(\sum_{i=1}^{m-1} p_i r^i \right). \end{aligned}$$

Hence,

$$r^m \neq \sum_{i=1}^{m-1} p_i r^i. \quad (1.9)$$

Thus r is not integral over $K_{v \leq 1}$, in particular, $K_{v \leq 1}$ is integrally closed.

For $K_{v < 1}$ being prime, obviously $1 \notin K_{v < 1}$. Let $a, b \in K_{v \leq 1} \setminus K_{v < 1}$. Then $v(a) = v(b) = 1$. Therefore $v(ab) = 1$. Showing $K_{v < 1}$ is a prime ideal. $\square$

Proposition 1.39. Let s be a specialization of $\eta \in \text{Spr}(R)$. We have,

$$a \in R \setminus \text{supp } s \Rightarrow K_{\eta < a} \subseteq K_{s < a}.$$

Applying this to $a = 1$, (recall that $\text{supp } s$ is a prime ideal), we get,

$$K_{\eta < 1} \subseteq K_{s < 1}.$$

Proof. By the assumption and Proposition 1.37, we know that $|a|_\eta \neq 0$. Therefore,

$$b \notin K_{\eta < a} \iff \eta \notin \mathcal{R}(b|a) \Rightarrow s \notin \mathcal{R}(b|a) \iff b \notin K_{s < a}.$$

$\square$

Proposition 1.40. Let s be a specilization of $\eta \in \text{Spr}(R)$.

$$a \in R \setminus \text{supp}(s) \Rightarrow K_{s \leq a} \subseteq K_{\eta \leq a}.$$

In particular for $a = 1$, we get,

$$K_{s \leq 1} \subseteq K_{\eta \leq 1}.$$

Proof. By Proposition 1.37, $|a|_\eta \neq 0$. Hence,

$$b \in K_{s \leq a} \iff s \in \mathcal{R}(a|b) \Rightarrow \eta \in \mathcal{R}(a|b) \iff b \in K_{\eta \leq a}.$$

$\square$

Definition 1.43. Let X be a linearly ordered set. For $a, b \in X$ with $a \leq b$, we define the line segment between a and b to be

$$|x, y|_X := \{x \in X \mid a \leq x \leq b\}.$$

A subset $C \subseteq X$ is convex if for any two elements $c, d \in C$, $c \leq d$, we have $|c, d|_X \subseteq C$.

Definition 1.44. A convex subgroup Θ is a subgroup of a linearly ordered group Γ which is convex.

Proposition 1.41. A subgroup Θ of a linearly ordered group Γ is convex if and only if for any $\theta \in \Theta$ with $\theta < 1$, $|\theta, \theta^{-1}|_\Gamma \subseteq \Theta$.

Recall that for a valuation $v : R \rightarrow \Gamma \cup \{0\}$, Γ_v is the subgroup of Γ generated by $v(R) \setminus \{0\}$.

Definition 1.45. $c\Gamma_v$ is the convex subgroup of Γ_v generated by $\{v(r) \mid r \in R, v(r) \geq 1\}$.

Remark 1.26. Equivalently, we can define $c\Gamma_v$ to be a subgroup generated by $\{v(r) \mid r \in R, v(r) > 1\}$.

Proposition 1.42.

$$c\Gamma_v = \{\gamma \in \Gamma_v \mid \exists r \in R \text{ s.t. } \gamma v(r) \geq 1\}$$

Remark 1.27. If H is a convex subgroup of Γ_v , then $c\Gamma_v \subseteq H$. Furthermore, we have,

$$v|_H(r) = \begin{cases} v(r) & (v(r) \in c\Gamma_v) \\ 0 & (\text{otherwise}). \end{cases}$$